they travel through the crystal and they emerge with a phase difference of 180°. When the two waves combine, they yield a plane-polarized wave, which has its plane of polarization rotated through an angle of 2θ.

Therefore, ***a half-wave plate rotates the plane of polarization of the incident plane polarized light through an angle 2θ.*** The half wave plate will invert the handedness of elliptical or circular polarized light, changing right to left and vice versa.

Now we are in a position to understand what happens when e-ray and o- ray overlap on each other after emerging from an anisotropic crystal plate. It is obvious that they cannot produce interference fringes as in a double slit experiment. On the other hand, they combine to produce different states of polarization depending upon their optical path difference.

1. When the optical path difference is 0 or an even or odd multiple of λ/2, the resultant light wave is linearly polarized.
2. When the optical path difference is λ/4, the resultant light wave is **elliptically polarized.**
3. In the particular instance when the wave amplitudes are equal and the optical path difference is λ/4, the resultant light wave is **circularly polarized**.

20.20. PRODUCTION OF ELLIPTICALLY POLARIZED LIGHT

A quarter wave plate and a polarizer are the optical devices necessary to produce elliptically polarized light from unpolarized light.

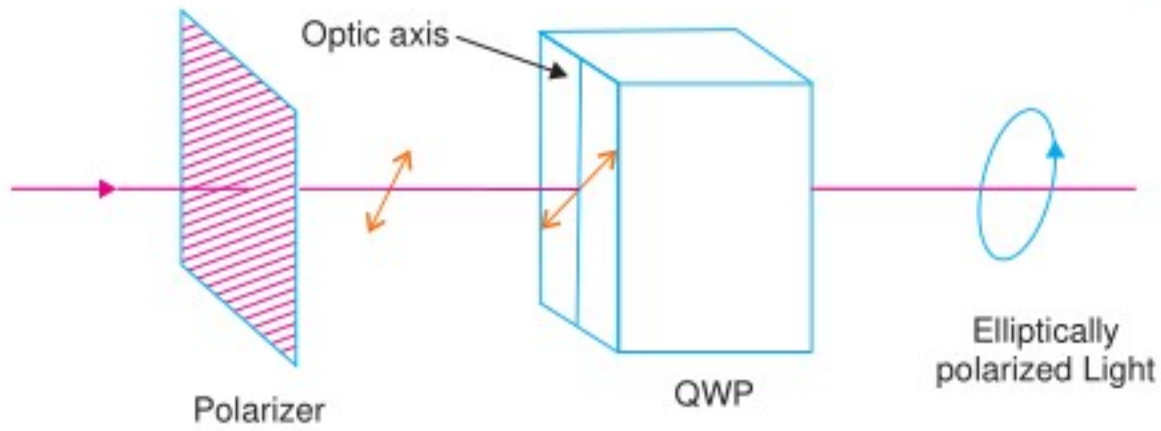


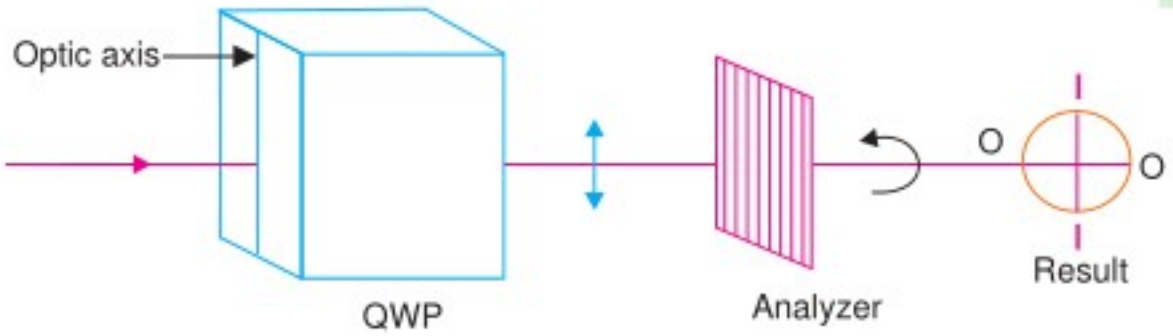
Fig. 20.47

Unpolarized light is first converted to plane polarized light by allowing it to pass through a polarizer (a polaroid sheet or a Nicol prism), as shown in Fig.20.47. The plane-polarized light is then made incident on a quarter wave plate. The quarter wave plate or the polarizer is rotated such that the electric vector E of plane polarized light wave makes an angle θ (≠45°) with the optic axis of the quarter wave plate. The incident ray divides into o-ray and e-ray of amplitudes E sin θ and E cos θ. The rays travel along the same direction in the crystal with different velocities. The two rays are polarized in orthogonal planes. They are in phase at the front face but progressively get out of phase as they travel through the crystal. When they emerge out of the crystal they will have a path difference of λ /4 or a phase difference of 90°. When they combine, they produce elliptically polarized light.

20.20.1 DETECTION OF ELLIPTICALLY POLARIZED LIGHT

The light beam is allowed to pass through an analyzer (a polaroid sheet or a Nicol prism). If on rotating the analyzing Polaroid sheet or Nicol, the intensity of the emerging beam varies from a maximum to a minimum value, but never reaching zero, then the incident light is elliptically polarized.

A similar result would be obtained if the incident light were partially polarized. The two cases may be distinguished by inserting a quarter wave plate in the path of light before it falls on the analyzer.



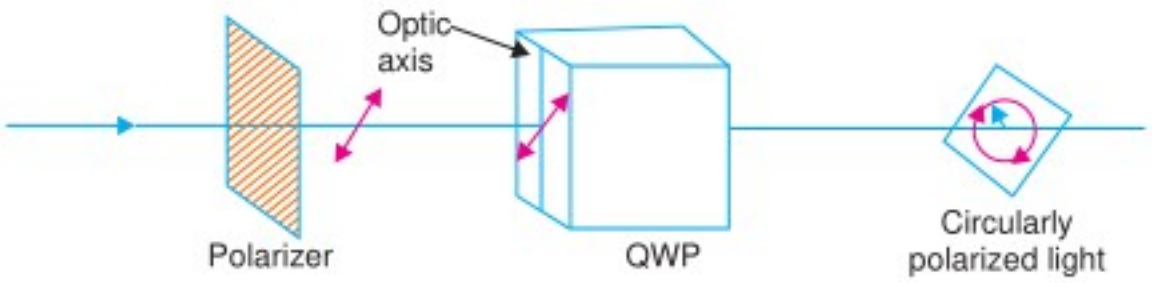
Detection of elliptically polarized light

Fig. 20.48

If the original light is elliptically polarized, it may be considered as resultant of two coherent plane polarized waves that is e-ray and o-ray, which are out of phase by 90°. If the light passes through the quarter wave plate, an additional phase difference of 90° is introduced between the e-ray and o-ray. Therefore, the total phase difference becomes 180° between the e-ray and o-ray. On emerging from the quarter plate, the e-and o-rays combine to produce plane-polarized light. If the light coming out of quarter wave plate is examined with an analyzer, light will be extinguished twice in one full rotation of the polarizer as shown in Fig.20.48.

20.21. PRODUCTION OF CIRCULARLY POLARIZED LIGHT

A quarter wave plate and a polarizer are the optical devices required for producing circularly polarized light from unpolarized light.



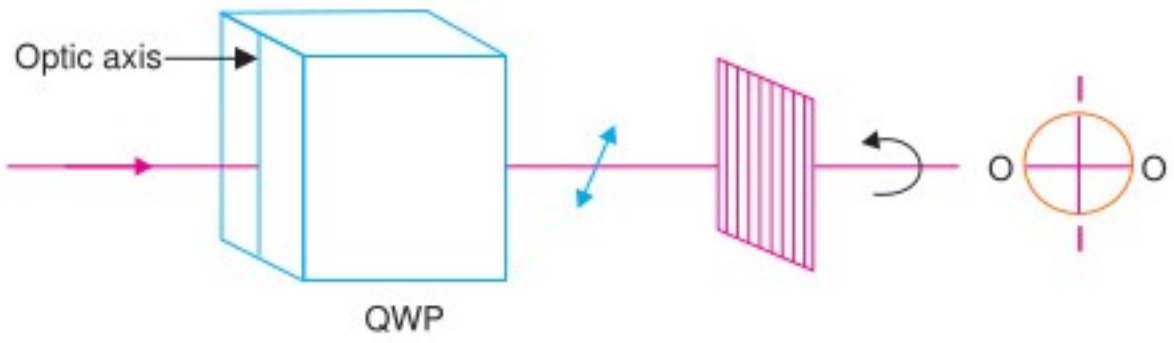
Production of Circularly Polarized Light

Fig. 20.49

Unpolarized light is first converted to plane polarized light by allowing it to pass through a polarizer (a polaroid sheet or a Nicol prism), as shown in Fig.20.49. Plane polarized light is then made incident on a quarter wave plate. The polarizer and the quarter wave plate are rotated such that the electric vector E of the plane-polarized wave makes an angle of 45° with the optic axis of the quarter wave plate. The plane polarized wave incident on the quarter wave plate splits into two rays, o-ray and e-ray of equal amplitude ($E_1 \cos 45° = E_2 \sin 45°$). The two rays travel in the same direction inside the crystal but with different velocities. The two rays are in phase at the front face of the crystal but progressively get out of phase as they travel through the crystal. As they emerge from the rear face of the crystal, they will have a path difference of $\lambda/4$ or phase difference of 90°. The two rays are linearly polarized in mutually perpendicular directions. When they combine, they produce circularly polarized light.

20.21.1 DETECTION OF CIRCULARLY POLARIZED LIGHT

The light beam is allowed to pass through an analyzer (a polaroid sheet or a Nicol prism). If on rotating the analyzing polaroid sheet or Nicol, the intensity of the emerging beam remains uniform, then the incident light is circularly polarized. A similar result would be obtained if the incident light is ordinary unpolarized light. The two cases may be distinguished by inserting a quarter wave plate in the path of light before it falls on the analyzer. If the original light is circularly polarized, it may be considered as resultant of two coherent plane-polarized waves, that is e-ray and o-ray, which are out of phase by 90°. If the light passes through the quarter wave plate, an additional phase difference of 90° is introduced between the e-ray and o-ray. Therefore, the total phase difference becomes 180° between the e-ray and o-ray. On emerging from the quarter plate, the e- and o-rays combine to produce plane-polarized light. Therefore, if the light coming out of quarter wave plate is examined with an analyzer, light will be extinguished twice in one full rotation of the polarizer as shown in Fig.20.50 or otherwise the incident light is unpolarized.



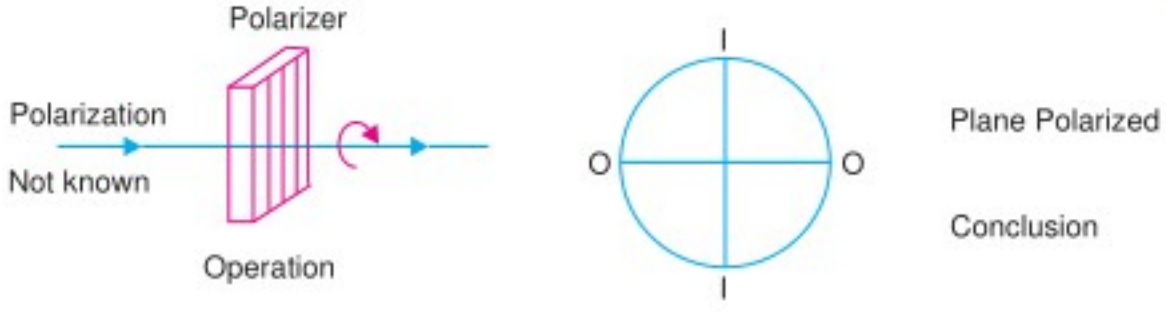
Detection of circularly polarized light

Fig. 20.50

20.22. ANALYSIS OF POLARIZED LIGHT

In practice light may exhibit any one of the three types of polarization, or may be unpolarized or a mixed type. The unaided eye cannot distinguish the different types of polarization. However, using a polarizer and a quarter wave plate, the actual type of polarization of a light beam can be ascertained. The following steps are used in the analysis of the type of polarization.

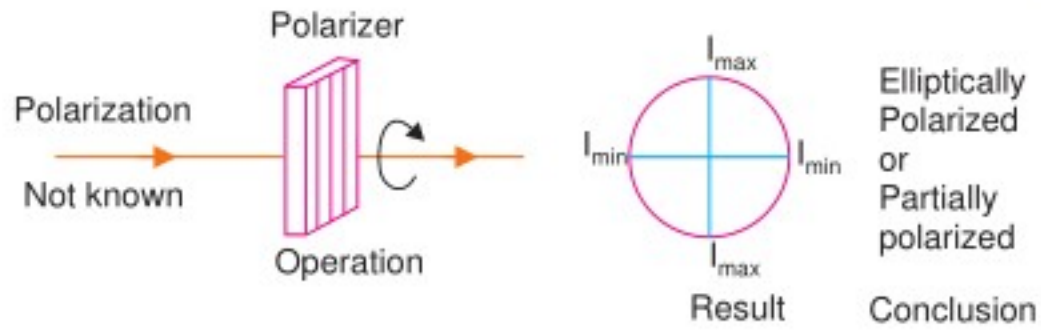
The light of unknown polarization is allowed to fall normally on a polarizer. The polarizer is slowly rotated through a full circle and the intensity of the transmitted light is observed. If the intensity of the transmitted light is extinguished twice in one full rotation of the polarizer, then the incident light is *plane polarized.*



(*a*)

Fig. 20.51

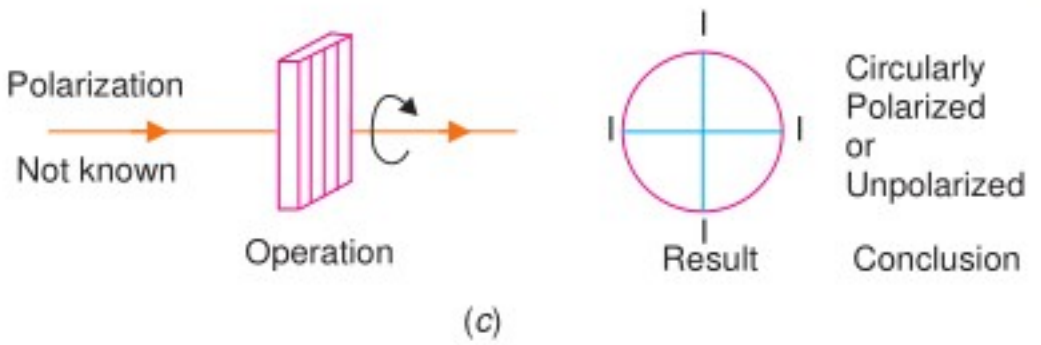
(*i*) If the intensity of the transmitted light varies between a maximum and a minimum value but does not become extinguished in any position of the polarizer, then the incident light is either elliptically polarized or partially polarized.



(*b*)

Fig. 20.51

(*ii*) If the intensity of the transmitted light remains constant on rotation of the polarizer, then the incident light is either circularly polarized or unpolarized.

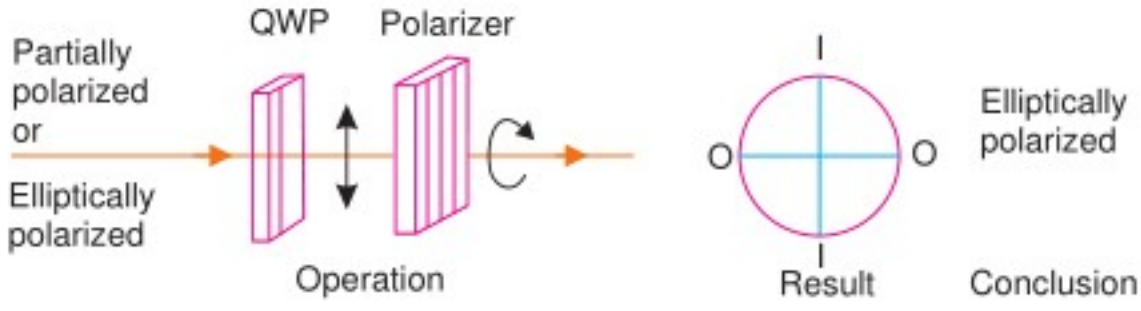


(*c*)

Fig. 20.51

To distinguish between elliptically polarized and partially polarized or between the circularly polarized and unpolarized light, we take the help of quarter wave plate. The light is first made to be incident on the quarter wave plate and then it passes through the polarizer.

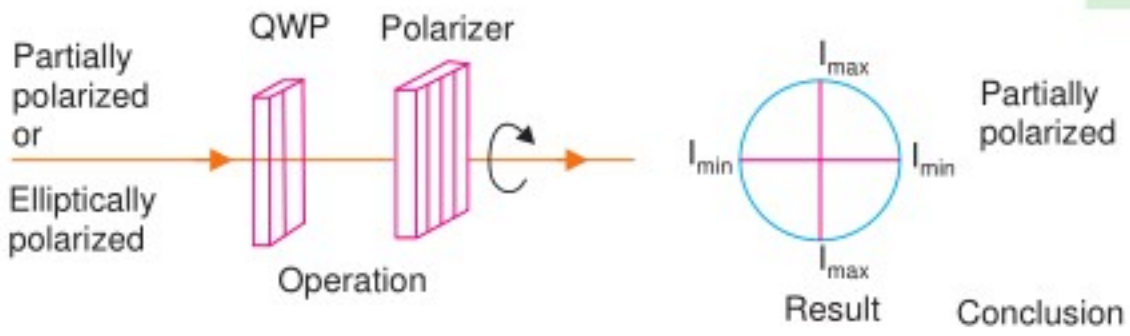
(*iii*) If the incident light is elliptically polarized, the quarter wave plate converts it into a plane polarized beam. When this polarized light passes through the polarizer, it would be extinguished twice in one full rotation of the polarizer.



(*d*)

Fig. 20.51

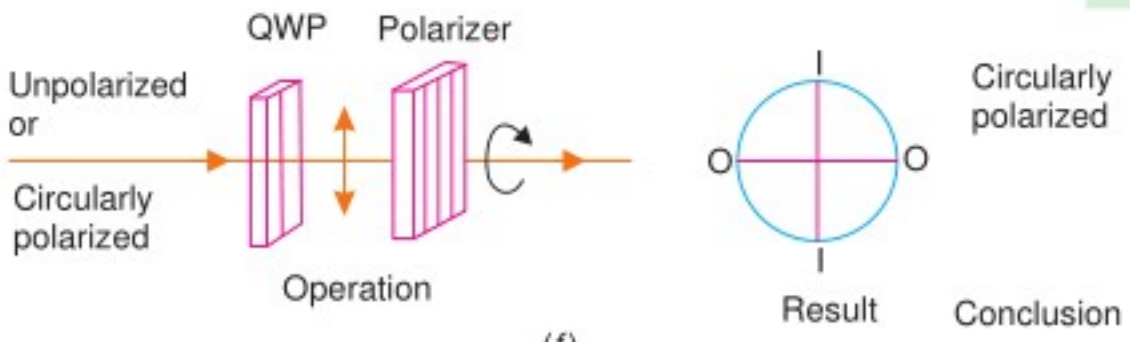
On the other hand, if the transmitted light intensity varies between a maximum and a minimum without becoming zero, then the incident light is partially polarized.



(*e*)

Fig. 20.51

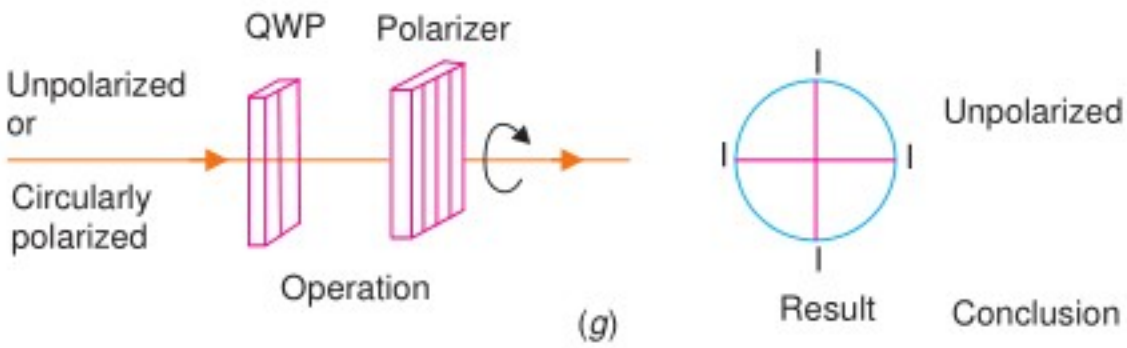
(*iv*) If the incident light is circularly polarized, the quarter wave plate converts it into plane polarized light. When this plane polarized light passes through the polarizer, it would be completely extinguished twice in one full rotation of the polarizer.



(*f*)

Fig. 20.51

On the other hand, if the intensity of the transmitted light stays constant, then the incident light is unpolarized.



(*g*)

Fig. 20.51

20.23. BABINET COMPENSATOR

A *compensator* is an optical device whose function is to compensate a path difference. It is used in conjunction with a polarizer and analyzer combination to investigate elliptically polarized light. The compensator helps in determining the axis of the ellipse and the ratio of their lengths. Elliptically polarized light may be considered as the resultant of two coherent plane polarized vibrations occurring in mutually orthogonal directions and with an initial path difference of $\lambda/4$. When the elliptically polarized light passes through the device such that it introduces a further path difference of $\lambda/4$. The total path difference between the perpendicular vibrations becomes $\lambda/2$ and the vibrations recombine after emerging from the device to produce plane polarized light. From the analysis of this plane polarized light, the information regarding the incident elliptically polarized light can be obtained.